

On an average SpaceShipTwo flight, the total time spent in sub-orbital space would be just a few minutes. And yes, inhabitants would experience zero-gravity in this! However, the total flight time from take-off to landing is expected to be around two and a half hours. Virgin Galactic has said it plans to start commercial spaceflights by 2013, although it hasn't confirmed that.

SpaceX

Scaled Composites and Virgin Galactic are focussed more on a 'luxury' space-flight. But the company that's at the forefront of emulating what national space agencies have managed is SpaceX, which has already successfully launched rockets (called the Falcon series) and even developed a spacecraft (called the Dragon) for manned missions which NASA itself has approved of.

In June 2010, the company successfully launched the Falcon 9 rocket, achieving Earth orbit. The launch vehicle is powered by a cluster of nine SpaceX-designed and developed Merlin engines. Using ultra pure jet fuel and liquid oxygen, the engines generated nearly a million pounds of thrust for the vehicle upon liftoff.

The Merlin engine is an orbit-class rocket engine and is the highest efficiency American hydrocarbon engine ever built. The Falcon 9 first stage, with a fully fuelled to dry weight ratio of over 20, has the world's best structural efficiency, despite being designed to higher human rated factors of safety.

The Dragon is a free-flying, reusable spacecraft made up of a pressurised capsule and unpressurised trunk used for Earth to LEO transport of pressurized cargo, unpressurized cargo, and/or crew members.



The Dragon spacecraft in orbit

It weighs about 6000 kilograms with the payload, with a volume of 10 cubic metres in the pressurised area and 14 cubic metres in the unpressurised area. Dragon can support up to seven passengers and is designed for a water landing, much like the Soyuz spacecraft.